

Youssef Ashraf

LinkedIn | GitHub | yousefashraf970@gmail.com | (+20) 01154202976

SUMMARY

Computer science professional passionate about deep learning and machine learning, with expertise in engineering AI-powered solutions. Experienced in building conversational AI systems, NLP pipelines, and deep learning models. Eager to contribute to innovative data-driven projects and continue deepening expertise in AI and data analysis.

EDUCATION

BNU (Benha National University) — Cairo, Egypt

B.Sc. in Computer Science, Major: Artificial Intelligence | Oct 2022 – July 2026 | GPA: 3.76

Graduated: July 2026

Relevant Coursework: Machine Learning · Deep Learning · Natural Language Processing · Neural Networks · Data Structures & Algorithms · Database Programming · Data Engineering & Visualization · Software Engineering · Statistics · Computer Networks

WORK EXPERIENCE

Cloud Services Trainee

2025

National Telecommunication Institute (NTI)

- Hands-on with 8 core AWS services: EC2, S3, IAM, VPC, Lambda, RDS, CloudWatch.
- Designed secure, scalable architectures aligned with Solutions Architect best practices.
- Reduced projected cloud costs by 20% through optimized IAM policies and networking.

Computer Vision Intern

2025

National Academy of IT for Persons with Disabilities (NAID) — New Capital · On-Site

- Contributed to a computer vision project for dyslexia detection; processed and analyzed 500+ text samples to improve reading assistance accuracy.
- Collaborated with a team of 6 researchers; contributed to 2 technical presentations on AI applications in accessibility, achieving 85% model accuracy in preliminary testing.

PROJECTS

BNUConnect – AI-Powered Academic Advisor Chatbot — Graduation Project

2026

- Engineered a production-grade AI academic advisor chatbot for Benha National University, deployed as a REST API (FastAPI + Uvicorn) integrated into the BNUConnect mobile app, serving students across 3 program tracks (AI & ML, Software Development, Data Science).
- Built a stateful LangGraph ReAct agent with a custom Judging Loop, orchestrating 12 LangChain tools across Neo4j knowledge graph (course catalog, prerequisites, program tracks), Supabase (student profiles, chat history), and a Pinecone RAG pipeline for university bylaw Q&A — achieving accurate multi-hop reasoning over the full academic domain.
- Designed a 5-step NLP preprocessing pipeline (reference resolution, entity extraction, fuzzy course name mapping, track normalization, query rewriting) that resolves abbreviations and pronouns before the agent runs, eliminating wasted tool calls and handling ambiguous queries via interactive disambiguation sessions.
- Implemented a RAG pipeline using HuggingFace multilingual-e5-large embeddings and Pinecone vector store (bnu-bylaws index) for bylaw/regulation Q&A, and built an interactive multi-turn course planner bridged from a CLI function to the chatbot via a thread I/O bridge with 90-second timeout handling.
- Stack: LangGraph, LangChain, FastAPI, Groq, Neo4j Aura, Supabase (PostgreSQL), Pinecone, HuggingFace, Python, Docker.

Auto Correction System

2025

- Designed and implemented a natural language processing system to automatically correct spelling and grammar errors using Transformer models and custom tokenization strategies.

Q&A Chatbot

2025

- Developed a question-answering chatbot capable of understanding user queries and providing context-aware responses using NLP techniques, with focus on intent recognition, entity extraction, and response generation via pre-trained Transformer models.

Facial Recognition Model with CNN

2024

- Developed a deep learning model using Convolutional Neural Networks (CNNs) for facial recognition and identity verification, handling image preprocessing and feature extraction.

Animal Segmentation Model with U-Net & TensorFlow

2024

- Built an animal segmentation model using the U-Net architecture with TensorFlow for pixel-wise classification. Involved preprocessing annotated datasets and evaluating performance with IoU and Dice coefficient metrics.

CERTIFICATIONS

- Machine Learning Specialization — Andrew Ng (Coursera)
- Deep Learning Specialization — Andrew Ng (Coursera)
- Hugging Face Agents Course

TECHNICAL SKILLS

Programming Languages: Python, C++

AI / ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn

LLM / Agent Frameworks: LangChain, LangGraph, Smolagents, LlamaIndex, LangFuse

Databases & Vector Stores: Neo4j, Supabase (PostgreSQL), Pinecone

APIs & Tools: FastAPI, Groq, Gradio, Weights & Biases, GitHub

Deep Learning Techniques: CNNs, RNNs, LSTMs, Transformers, Transfer Learning, LLMs

Data Science Skills: Statistical Analysis, Data Visualization, Feature Engineering